



ITAI 1378 •
Midterm
Blueprint

PlateVision

AI-Assisted Vehicle & License Plate
Analysis from Security Footage

Project Team

Tier 2 — Advanced

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Motivation: A Real-World Problem

From a real incident to a CV question

Police contacted a family while looking for information about a black Corvette seen on their street. Ring footage captured the vehicle, but finding a clear view of the plate across many frames was difficult. This inspired our question: Can computer vision help identify and organize the clearest available vehicle and plate evidence from security footage?

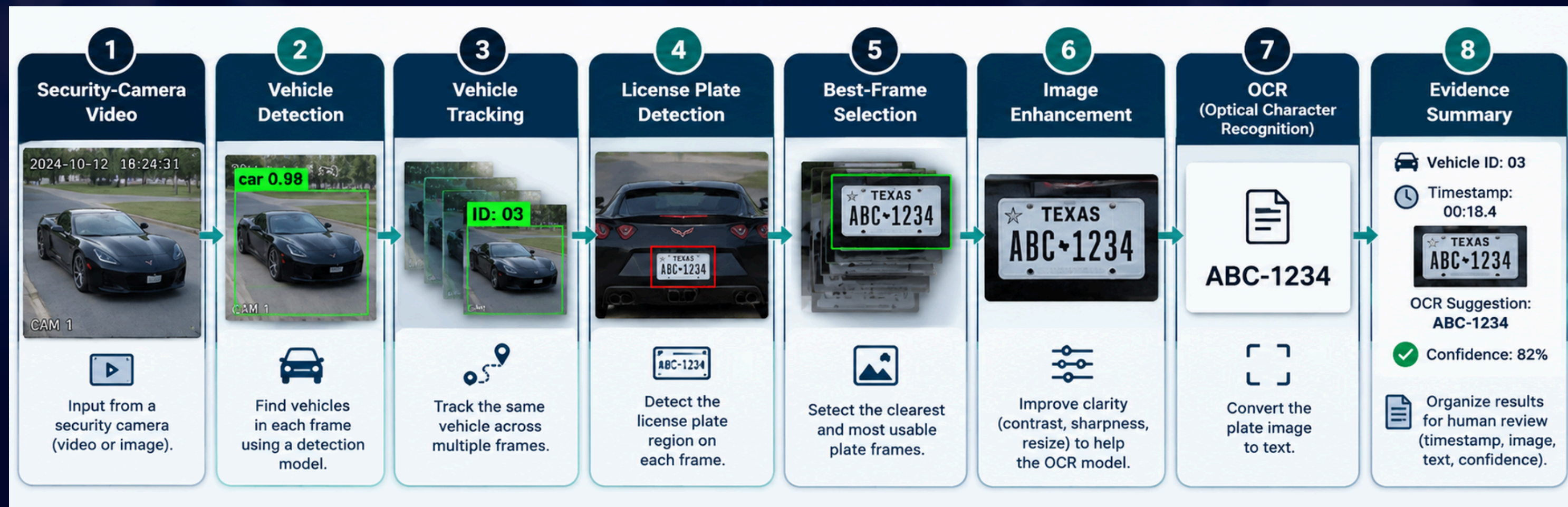
- Who cares: people reviewing residential or business security footage, including homeowners, security personnel and investigators.





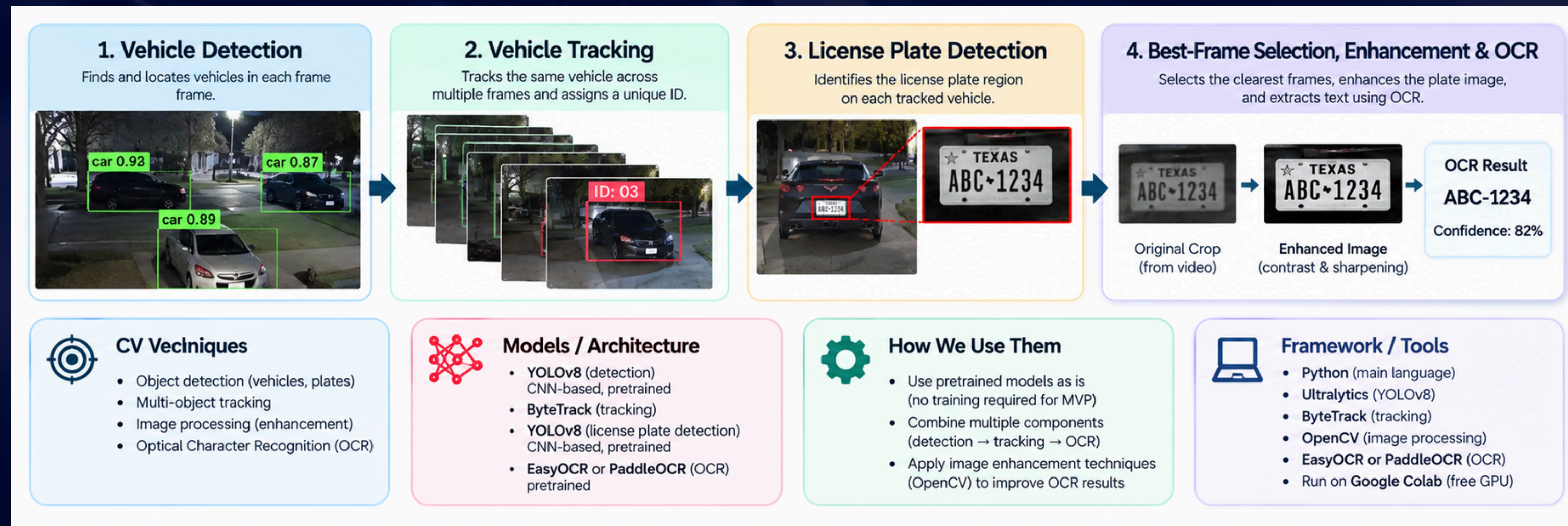
From Problem to Proposed Solution

PlateVision analyzes security-camera video to locate vehicles and license plates, select the clearest available plate frames, and organize the results for human review.





How PlateVision Works





Data Strategy

Primary Dataset — UFPR-ALPR

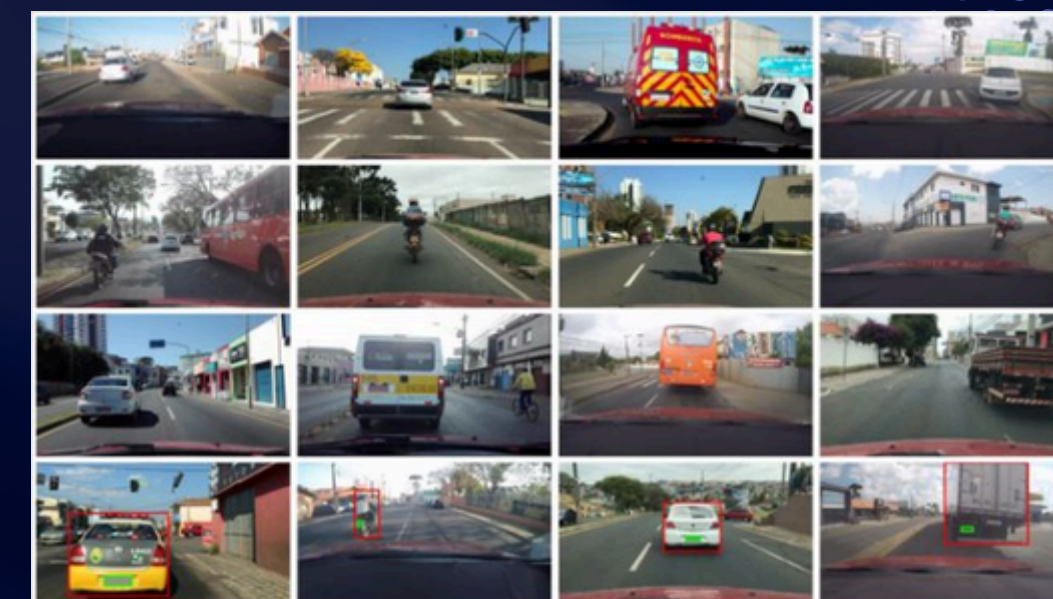
- 4,500 annotated images
- 150 vehicles
- 1920 × 1080 resolution
- Real-world vehicle scenarios
- Existing vehicle, license-plate, and character annotations

How We Will Use It

- License-plate detection/testing
- Evaluate plate localization
- Test OCR on visible plates
- Use only a manageable subset initially for the First Working Demo

Real-World Testing

- A small number of security-camera video samples
- Used to test the complete PlateVision pipeline
- Not used as the primary training dataset
- No personally identifiable footage shared publicly



UFPR-ALPR Dataset

This dataset, called UFPR-ALPR dataset, includes 4,500 fully annotated images (over 30,000 LP characters) from 150 vehicles in real-world scenarios where both the vehicle and the camera (inside...

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UFPR-ALPR Dataset License Agreement

The UFPR-ALPR dataset is released for academic research only and is free to researchers from educational ...

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Access requested



Success Metrics

LICENSE PLATE DETECTION

≥ 0.70 mAP@50

**Correctly
detects and
localizes license
plates**

OCR ACCURACY

$\geq 70\%$

**Correctly reads
complete visible
plates**

USABLE FRAME

$\geq 80\%$

**Detected
vehicles produce
a usable plate
frame**



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Milestone Plan

WEEK 10



1. BLUEPRINT

Finalize project plan,
dataset, workflow &
metrics.

WEEK 11



2. FIRST WORKING DEMO

Run pretrained
detection on sample
images/video.

WEEKS 12-13



3. MAKE IT YOURS

Integrate vehicle
tracking, plate
detection,
enhancement
& OCR.

WEEK 14



4. IMPROVE & MEASURE

Test the system,
calculate metrics &
document failure cases.

WEEK 15



5. PACKAGE & PRESENT

Complete demo,
README, final slides
& presentation.



Risks & Resources

⚠️ RISK 1 — Poor Video Quality

Plates may be too small, blurry, dark, or partially visible.

Plan B

Focus on vehicle detection, tracking, and best-frame selection. Flag unreadable plates instead of forcing an OCR result.

⚠️ RISK 2 — Inconsistent OCR Results

OCR may produce different plate readings across video frames.

Plan B

Compare OCR results from multiple high-quality frames and present the best candidate with confidence information for human verification.



Resources

Compute

Google Colab

Models/Tools

YOLO • ByteTrack • OpenCV • OCR

Dataset

UFPR-ALPR
access requested

Estimated Cost

\$0

All project resources will use free/open-source or academic-use options.